

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Francium[-223]		1		1		
		(ii)		60 [hours] in Sodium-24 column (1) [Technetium-99m column] 6 [hours] (1) 12-18-24-30 [hours] (1)		3		3	2	
	(b)	(i)		<u>Gamma</u> passes through steel or is very penetrating (1) The reading will rise when passing through a crack (1) Accept isotope decays slowly or iridium has a long half-life or it is active for a long time		2		2		
		(ii)		<u>Beta</u> {is ionising / kills [tumour] cells / absorbed by tumour / can penetrate the [whole] tumour} (1) Isotope remains active for long enough (1)		2		2		
		(iii)	I	Emits alpha radiation (1) which ionises [cells] the most or half-life is quite long or takes 3 weeks to decay to safe level (1)	1	1		2		
			II	Depends on [type of] rock Accept it increases with an earthquake	1			1		
				Question 5 total	2	9	0	11	2	0